Anmol Goyal

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EDUCATION

SRM Institute of Science and Technology (SRMIST), Kattankulathur 2025 – 2029
B.Tech in Computer Science and Engineering Chennai, India

SKILLS

Languages : C, C++
Domains : Systems programming, graphics/rendering, parallel computing (OpenMP)
Tools : Git, Make, Neovim, Linux

PROJECTS

Ray Tracer C++

- Built a CPU ray tracer from scratch in C++ (following *Ray Tracing in One Weekend*), implementing ray–sphere intersection, recursive path tracing, and physically based materials (diffuse, metal, dielectric)
- Implemented antialiasing via stochastic sampling, defocus-blur camera model, and gamma correction; extended the renderer with a custom Saturn-in-space scene by reworking the sky-gradient shading
- Reasoned about the underlying vector math (reflection/refraction, Snell's law, Lambertian scattering) rather than treating the pipeline as a black box

Mini Compiler C++, x86-64 Assembly

- Built a compiler front-to-back for a minimal C subset: lexer → recursive descent parser → typed AST → x86-64 (AT&T syntax) code generation, with no external compiler frameworks
- Verified the full pipeline end to end: source compiles to native assembly that assembles, links, and returns the correct exit code

Matrix-OpenMP-Benchmark C, OpenMP

- Parallelized 1000×1000 integer matrix multiplication ($\sim 10^9$ multiply-add operations, $O(N^3)$) with OpenMP, parallelizing the outer loop via `#pragma omp parallel for`
- Benchmarked against a serial baseline using `omp_get_wtime()`, measuring a **3.8x speedup** (4.25s → 1.12s) and analyzing the gap from linear scaling as a memory-bandwidth limit